Michael Fröwis

Chief Technology Officer (CTO), Iknaio Cryptoasset Analytics GmbH
PhD in Computer Science (Security & Privacy Lab), University of Innsbruck

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Computer scientist and CTO of Iknaio, working at the intersection of cryptoasset forensics, program analysis, and production-grade data systems. PhD research on the large-scale analysis of smart contracts (University of Innsbruck, 2022); since then responsible for the technology behind Iknaio's investigation products for law enforcement, regulators, and financial institutions. Contributor to the national and EU research projects TITANIUM and KRYPTOMONITOR, co-author of 9 peer-reviewed papers on Ethereum, DeFi, and forensic cryptoasset investigations, and trainer of Austrian law-enforcement and tax authorities. Brings more than seven years of prior industry experience as a software developer in banking.

Key Achievements

- **Research-to-product transfer** – As CTO of Iknaio, leads the technical transfer of research results into the production analytics platform built on the open-source GraphSense stack, including cross-case deconfliction of investigations, attribution heuristics, broader cryptoasset coverage, and AI-assisted investigation workflows.
- **Smart-contract forensics** – Developed reverse-engineering and program-analysis methods for Ethereum at scale: control-flow immutability, token-system detection, `CREATE2` deployment patterns, and detection of privileged parties in contracts.
- **Evidence that holds in court** – In the EU H2020 project TITANIUM, bridged the technical and legal partners and co-authored the reference work on safeguarding the evidential value of forensic cryptocurrency investigations.
- **Capacity building for public authorities** – Organised and taught two cryptoasset-forensics trainings (Innsbruck 2021, Vienna 2022) for the Austrian law-enforcement and tax-authority partners of KRYPTOMONITOR.
- **DeFi transparency** – Co-authored work on the verifiability of Total Value Locked (IEEE ICBC 2025) and on decentralised exchanges, airdrops and governance tokens (*Journal of Corporate Finance*, 2023).
- **Industry and academia** – Combines a vocational IT apprenticeship and seven years of banking software development with an academic research career, reaching the PhD via the second-chance education path (Berufsreifeprüfung).

Professional Experience

2022 – Present	**Chief Technology Officer (CTO)**, *Iknaio Cryptoasset Analytics GmbH, Vienna*. Responsible for technology strategy, platform architecture, and engineering; turns research into investigation products and services. Joined as software architect.
2017 – 2022	**Research Assistant (PhD)**, *Security and Privacy Lab, University of Innsbruck*. Cryptoasset forensics and smart-contract analysis in TITANIUM and KRYPTOMONITOR; teaching.
2008 – 2012	**Software Developer**, *Hypo Landesbank Vorarlberg, Bregenz*.
2005 – 2008	**Apprenticeship as IT Technician**, *Hypo Landesbank Vorarlberg, Bregenz*.

Education

2022	**PhD in Computer Science**, *University of Innsbruck*. Thesis: *Analyzing a Smart Contract Platform at Scale*; advisor Prof. Rainer Böhme.
2017	**MSc in Computer Science**, *University of Innsbruck*. Thesis: *Analysis of Solidity Programming Pitfalls and Compile-time Solutions*.
2015	**BSc in Computer Science**, *University of Innsbruck*.
2012	**University entrance qualification** (Berufsreifeprüfung), *Volkshochschule Götzis*.

Selected Research Projects

2024 – 2027	**ENSEMBLE** (Horizon Europe CL3), *Task Leader, T3.5 Automated Cryptoasset Transaction Tracing Workflows*. Enhancing collaborative investigations against cross-border cybercrime with AI-based tools, joint investigation processes, and training; consortium of 18 partners including five law-enforcement agencies.
2020 – 2022	**KRYPTOMONITOR** (FFG KIRAS), *Researcher*. Forensic methods for smart contracts and off-chain transactions; reverse-engineering tools for smart contracts; organisation of two trainings for law enforcement and tax authorities.
2017 – 2020	**TITANIUM** (EU Horizon 2020), *Researcher*. Tools for investigating transactions in underground markets; liaison between legal and technical partners on court-proof evidence reports.

Selected Talks, Trainings & Teaching

2022	**Cryptoasset Forensics Training 2**, KRYPTOMONITOR, Vienna. Revised second edition for law-enforcement and tax-authority partners.
2021	**Cryptoasset Forensics Training 1**, KRYPTOMONITOR, Innsbruck. Lectures on cryptoasset systems and transaction tracing in Ethereum; hands-on tracing lab with GraphSense.
2019	**Proseminar Computer Architecture**, University of Innsbruck. Two first-semester groups, from digital circuit design to ARM assembly.
2018	**Tracking Payment Flows in Ethereum**, Symposium on Post-Bitcoin Cryptocurrencies, AIT, Vienna.

Selected Publications

1. Saggese, P., **Fröwis, M.**, Kitzler, S., Haslhofer, B., & Auer, R. (2025). Towards verifiability of total value locked (TVL) in decentralized finance. In *2025 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)* (pp. 1–9). IEEE. https://doi.org/10.1109/ICBC64466.2025.11114539

2. Haslhofer, B., Hanslbauer, C., **Fröwis, M.**, & Goger, T. (2023). Increasing the efficiency of cryptoasset investigations by connecting the cases. In *2023 APWG Symposium on Electronic Crime Research (eCrime)* (pp. 1–10). IEEE. https://doi.org/10.1109/eCrime61234.2023.10485550

3. **Fröwis, M.**, & Böhme, R. (2023). Detecting privileged parties on Ethereum. In *Financial Cryptography and Data Security. FC 2023 International Workshops* (pp. 470–488). Springer. https://doi.org/10.1007/978-3-031-48806-1_30

4. Makridis, C. A., **Fröwis, M.**, Sridhar, K., & Böhme, R. (2023). The rise of decentralized cryptocurrency exchanges: Evaluating the role of airdrops and governance tokens. *Journal of Corporate Finance*, *79*, 102358. https://doi.org/10.1016/j.jcorpfin.2023.102358

5. **Fröwis, M.**, Gottschalk, T., Haslhofer, B., Rückert, C., & Pesch, P. (2020). Safeguarding the evidential value of forensic cryptocurrency investigations. *Forensic Science International: Digital Investigation*, *33*, 200902. https://doi.org/10.1016/j.fsidi.2019.200902

6. **Fröwis, M.**, Fuchs, A., & Böhme, R. (2019). Detecting token systems on Ethereum. In *Financial Cryptography and Data Security (FC 2019)* (pp. 93–112). Springer. https://doi.org/10.1007/978-3-030-32101-7_7

7. **Fröwis, M.**, & Böhme, R. (2017). In code we trust? Measuring the control flow immutability of all smart contracts deployed in Ethereum. In *Data Privacy Management, Cryptocurrencies and Blockchain Technology (CBT 2017)* (pp. 357–372). Springer. https://doi.org/10.1007/978-3-319-67816-0_20